

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : TOAS, TNAS & TAAS)
ANSWER KEY
PAPER-1
PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6	7	8
	A.	A,C	A,C,D	B,C	B,D	B,D	A,C	B,C	A,D
SECTION-I (ii)	Q.	9	10	11	12				
	A.	A	B	D	B				
SECTION-II	Q.	1	2	3	4	5	6		
	A.	18.00	2.66 to 2.67	48.00	20.00	3.50	9.00		

PART-2 : CHEMISTRY

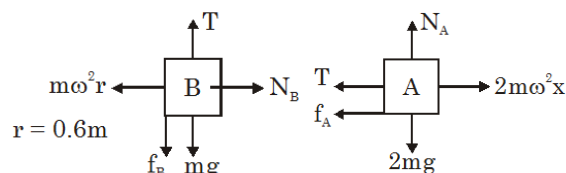
SECTION-I (i)	Q.	1	2	3	4	5	6	7	8
	A.	B,C,D	A,B,D	A,C	B,C	B,D	B,D	A,B,D	A,C
SECTION-I (ii)	Q.	9	10	11	12				
	A.	D	C	D	C				
SECTION-II	Q.	1	2	3	4	5	6		
	A.	4.00	5.00	3.00	9.60	240.00	2.00		

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4	5	6	7	8
	A.	A,B,C,D	A,B,C	A,C	A	B,C,D	B,C	A	A,B,C,D
SECTION-I (ii)	Q.	9	10	11	12				
	A.	D	A	B	D				
SECTION-II	Q.	1	2	3	4	5	6		
	A.	0.14	8.33	2.00	0.25	2100.00	0.00		

HINT – SHEET
PART-1 : PHYSICS
SECTION-I (i)

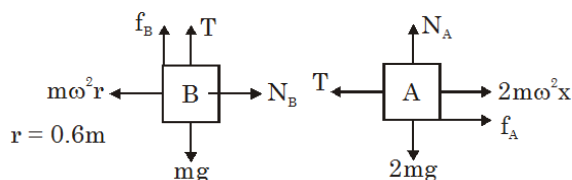
1. Ans (A,C)

 For $x_{\max} \Rightarrow$


$$2m\omega^2 x = T + f_A \quad \dots (i)$$

$$T = f_B + mg \quad \dots (ii)$$

$$f_A = \mu N_A, f_B = \mu N_B$$

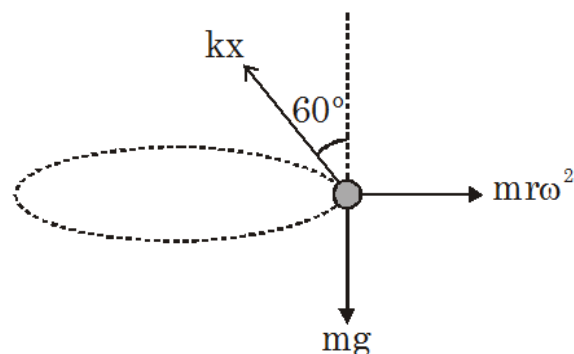
 For $x_{\min} \Rightarrow$


$$f_A + 2m\omega^2 x = T \quad \dots (i)$$

$$T + f_B = mg \quad \dots (ii)$$

$$f_A = \mu N_A, f_B = \mu N_B$$

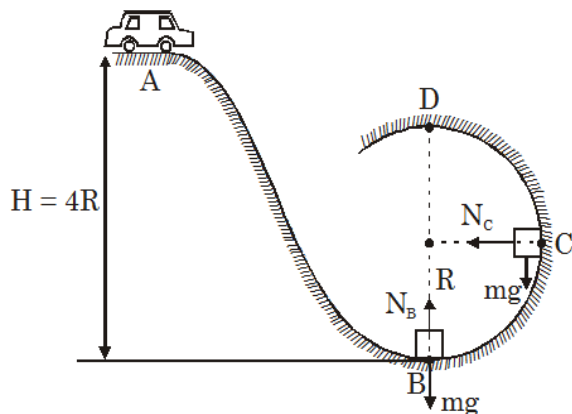
2. Ans (A,C,D)



$$kx \cdot \cos 60 = mg$$

$$kx \sin 60 = mr\omega^2$$

3. Ans (B,C)



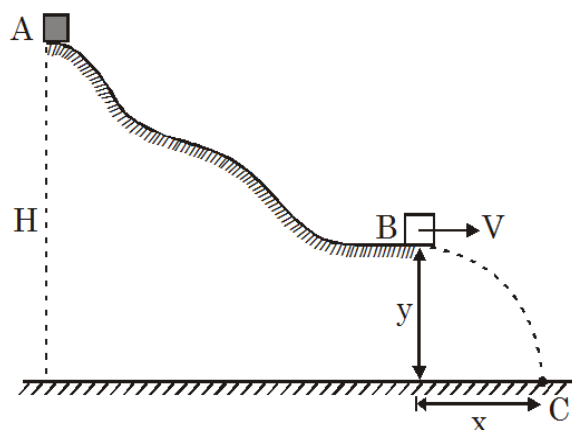
$$V_B = \sqrt{2gH} = \sqrt{8gR}$$

$$V_C = \sqrt{2g(3R)} = \sqrt{6gR}$$

$$N_B - mg = \frac{mV_B^2}{R}$$

$$N_C = \frac{mV_C^2}{R}$$

4. Ans (B,D)



$$mg(H - y) = \frac{1}{2}mv^2$$

$$v = \sqrt{2g(H - y)}$$

$$t = \sqrt{\frac{2y}{g}}$$

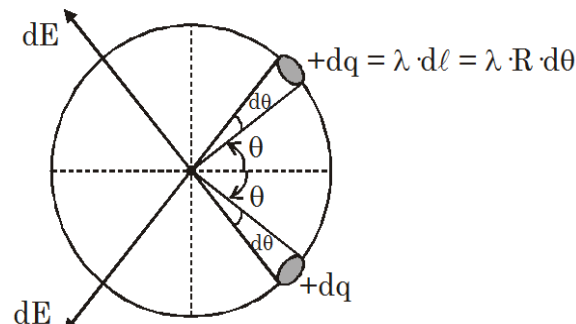
$$x = vt = 2\sqrt{y \cdot (H - y)}$$

$$\text{For } x_{\max} \Rightarrow \frac{dx}{dy} = 0$$

$$y = \frac{H}{2}$$

$$x_{\max} = H$$

5. Ans (B,D)

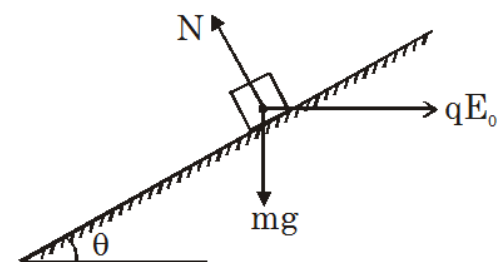


$$E = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} 2 \cdot dE \cos \theta$$

$$= 2 \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{K \cdot dq}{R^2} \cdot \cos \theta$$

$$= 2 \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{K \cdot \lambda_0 \cos \theta \cdot R \cdot d\theta \cdot \cos \theta}{R^2}$$

6. Ans (A,C)



$$N = mg \cos \theta + qE_0 \sin \theta$$

$$mg \sin \theta = qE_0 \cos \theta$$

7. Ans (B,C)

$$\frac{1}{10} = (1.5 - 1) \left(\frac{2}{R} \right)$$

$$R = 10 \text{ cm}$$

$$P = \frac{1.5 - 1}{10} + \frac{\frac{4}{3} - 1.5}{-10}$$

$$\frac{4}{3v} - \frac{1}{-21} = \frac{1}{20} + \frac{1}{60}$$

8. Ans (A,D)

$$\sin i_g = \mu \sin (A - \theta_c)$$

$$= \mu [\sin A \cos \theta_c - \cos A \sin \theta_c]$$

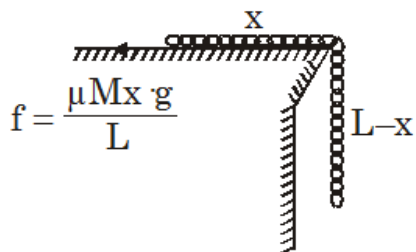
$$\sin i_g = \mu \left[\sin A \cdot \sqrt{1 - \frac{1}{\mu^2}} - \cos A \cdot \frac{1}{\mu} \right]$$

$$i_g = 30$$

PART-1 : PHYSICS

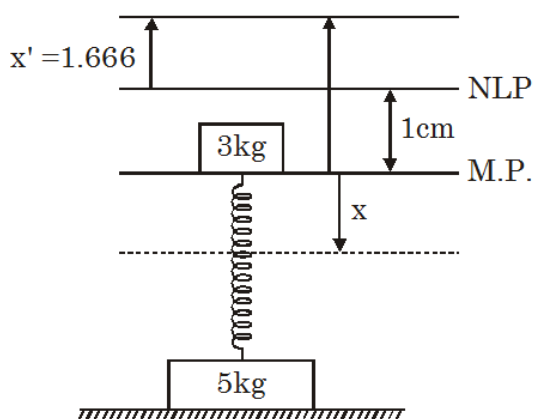
SECTION-II

1. Ans (18.00)



$$\omega = - \int_{\frac{L}{3}}^0 \frac{\mu M}{L} g x \cdot dx = \frac{-\mu M g L}{18}$$

2. Ans (2.66 to 2.67)



$$k \times \frac{1}{100} = 30$$

$$k = 3000$$

$$kx' = 50$$

$$x' = \frac{50}{3000}$$

$$= \frac{5}{3}$$

$$= 1.666 \text{ cm}$$

$$x = 2.666$$

3. Ans (48.00)

$$P = \left(\frac{\mu_A - 1}{\alpha} \right) \cdot 2 + \left(\frac{\mu_B - \mu_A}{-R} \right) \cdot 2 + \left(\frac{-2\mu_B}{-\frac{R}{2}} \right)$$

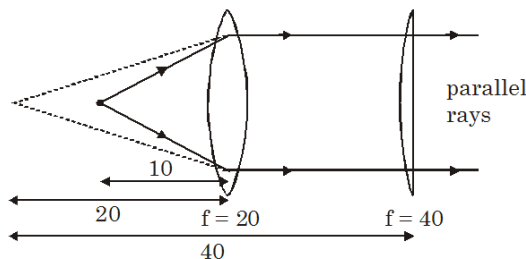
And

$$P = -\frac{1}{f}$$

$$\text{From } \frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

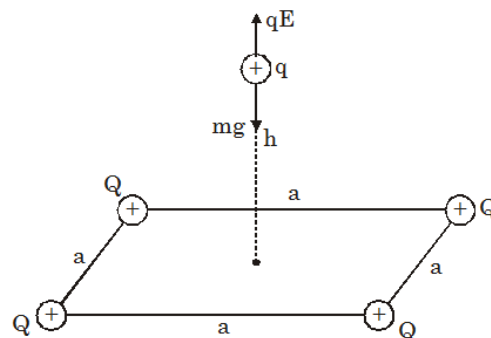
$$\frac{1}{-40} + \frac{1}{-10} = \frac{1}{f}$$

4. Ans (20.00)



$$d = 20 \text{ cm}$$

5. Ans (3.50)



$$qE = mg$$

$$\frac{q \cdot KQ \cdot h}{\left[h^2 + \left(\frac{a}{\sqrt{2}} \right)^2 \right]^{\frac{3}{2}}} = mg$$

6. Ans (9.00)

$$x = 3 \sin(100t) + 8 \cos^2(50t)$$

$$x = 3 \sin(100t) + 8 \left[\frac{\cos(100t) + 1}{2} \right]$$

$$= 4 + 3 \sin(100t) + 4 \cos(100t)$$

$$= 4 + 5 \sin(100t + 53^\circ)$$

$$x_{\max} = 9 \text{ cm}$$

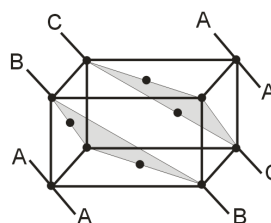
PART-2 : CHEMISTRY

SECTION-I (i)

4. Ans (B,C)

(A) SC involve 2-D square arrangement of atoms.

(B) Correct,



(C) Correct, HCP can also be formed by ABABABAB arrangements where A and B 2d hexagonal arrangements.

(D) False.

5. Ans (B,D)

$$C = \frac{600}{7}\% \Rightarrow \frac{600}{12 \times 7} = \frac{50}{7}$$

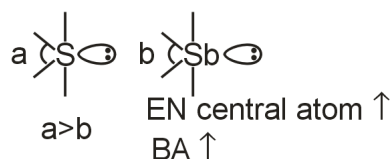
$$H = \frac{100}{7}\% \Rightarrow \frac{100}{1 \times 7} = \frac{50}{7}$$

$$\therefore \frac{C}{H} \text{ ratio} = \frac{1}{2} \Rightarrow \text{EF} = \text{CH}_2$$

$$\text{Molar mass} = 93.33 \times 10^{-24} \times 6 \times 10^{23}$$

$$\therefore \text{molecule formula} = \text{C}_4\text{H}_8$$

6. Ans (B,D)



PART-2 : CHEMISTRY

SECTION-I (ii)

9. Ans (D)

$$\text{By definition, } \alpha = \frac{C_{A_0} - C_A}{C_{A_0}}$$

$$\text{For zero order, } C_A = C_{A_0} - kt \Rightarrow \alpha = \frac{kt}{C_{A_0}}$$

$$\text{For 1st order, } C_A = C_{A_0} e^{-kt}$$

$$\Rightarrow C_{A_0}(1 - \alpha) = C_{A_0} e^{-kt}$$

$$\Rightarrow \alpha = 1 - e^{-kt}$$

$$\text{For 2nd order, } \frac{1}{C_A} = \frac{1}{C_{A_0}} + kt$$

$$\text{or, } \frac{C_{A_0} - C_A}{C_A C_{A_0}} = kt$$

$$\text{or, } \alpha = \frac{C_{A_0} kt}{1 + C_{A_0} kt}$$

PART-2 : CHEMISTRY

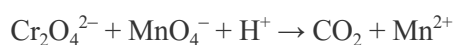
SECTION-II

4. Ans (9.60)

$$\pi = CRT = 0.4 \text{ M} \times 0.08 \frac{\text{L} \cdot \text{atm}}{\text{molK}} \times 300 \text{ K} = 9.6 \text{ atm}$$

5. Ans (240.00)

$$0.5 \text{ M KMnO}_4$$



$$\text{v.f.} = 2 \quad \text{v.f.} = 5, 0.5 \text{ M}$$

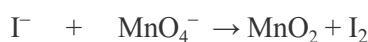
$$1.5 \text{ M, } V_1 \text{ ml}$$

$$125 \text{ M}$$

$$\text{eq. of MnO}_4^- = \text{eq. of C}_2\text{O}_4^{2-}$$

$$\text{or, } 0.5 \times V_1 \times 5 = 1.5 \times 125 \times 2$$

$$\text{or, } V_1 = 150 \text{ ml}$$



$$0.5 \text{ M, } 0.5 \text{ M}$$

$$270 \text{ ml}$$

$$\text{v.f.} = 1 \quad \text{v.f.} = 3$$

$$V_2 \text{ ml}$$

$$\text{eq. of I}^- = \text{eq. of MnO}_4^-$$

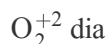
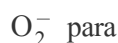
$$\text{or, } 0.5 \times 270 \times 1 = 0.5 \times V_2 \times 3$$

$$\text{or, } V_2 = 270 \text{ ml}$$

$$\therefore \text{Total volume of KMnO}_4 = V_1 + V_2 = 150 + 90 = 240 \text{ ml}$$

6. Ans (2.00)

s-p orbit mixing will be in 14 or urs proton system



PART-3 : MATHEMATICS

SECTION-I (i)

1. Ans (A,B,C,D)

$$\text{At } x=y=0: f(0)=0$$

$$\text{At } x=0; f(y)=f(-y) \Rightarrow f \text{ is even}$$

$$\text{replace } y \rightarrow -y:$$

$$f(x-y)-f(x+y)=f(x).f(-y)$$

$$\therefore f(x).f(y)=-f(x).f(-y)$$

$$\Rightarrow f(x)(f(y)+f(-y))=0$$

$$\Rightarrow f(x)=0 \text{ or } f(y)=-f(-y) (f \text{ is odd})$$

$$\therefore \text{either } f(x)=0 \text{ or } f(x) \text{ is simultaneously odd and even.}$$

$$\therefore f(x)=0 \text{ only.}$$

2. Ans (A,B,C)

Let $\angle B = \theta$

$$\therefore \angle C = 2\angle B = 2\theta$$

$$\angle A + \angle B + \angle C = 180^\circ$$

$$\angle A + \theta + 2\theta = 180^\circ$$

$$\angle A = 180^\circ - 3\theta$$

$$\frac{\angle A}{2} = 90^\circ - \frac{3\theta}{2}$$

Now in $\triangle ABD$

$$\frac{AD}{\sin \theta} = \frac{BD}{\sin(90^\circ - \frac{3\theta}{2})}$$

$$AD = \frac{BD \sin \theta}{\cos \frac{3\theta}{2}} \quad \dots(1)$$

$$\text{In } \triangle ADC \quad \frac{AD}{\sin 2\theta} = \frac{AC}{\sin(90^\circ - \frac{\theta}{2})}$$

$$\text{since } AC = BD \quad \dots(2)$$

From (1) and (2)

$$\frac{\sin \theta}{\cos \frac{3\theta}{2}} = \frac{\sin 2\theta}{\cos \frac{\theta}{2}}$$

$$\sin \theta \cos \frac{\theta}{2} = \sin 2\theta \cdot \cos \frac{3\theta}{2}$$

$$\sin \theta \cos \frac{\theta}{2} = 2 \sin \theta \cos \theta \cdot \cos \frac{3\theta}{2}$$

$$\cos \frac{\theta}{2} = \cos \frac{5\theta}{2} + \cos \frac{\theta}{2}$$

$$\cos \frac{5\theta}{2} = 0 \quad \therefore \frac{5\theta}{2} = 90^\circ$$

$$\theta = 36^\circ$$

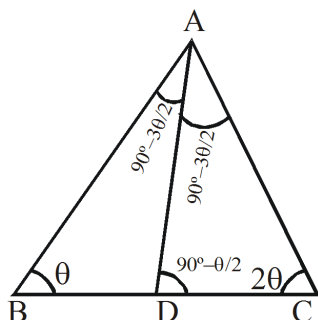
Hence

$$\angle B = 36^\circ, \angle C = 72^\circ$$

$$\angle A = 180^\circ - 3\theta = 180^\circ - 108^\circ = 72^\circ$$

$$\therefore \angle A = \angle C$$

Now check the options



3. Ans (A,C)

$$PA = x, PB = y, PC = z$$

Applying cosine laws :

$$\cosine \Rightarrow \begin{cases} x^2 = z^2 + b^2 - 2bz \cos \alpha \\ y^2 = x^2 + c^2 - 2cx \cos \alpha \\ z^2 = y^2 + a^2 - 2ay \cos \alpha \end{cases}$$

$$\text{Adding } 2(cx + ay + bz) \cos \alpha = a^2 + b^2 + c^2$$

$$\text{Also for } \triangle ABC \quad \Delta = \frac{1}{2} (cx + ay + bz) \sin \alpha$$

$$\tan \alpha = \frac{4\Delta}{a^2 + b^2 + c^2} = \frac{168}{295} \Rightarrow m + n = 463$$

4. Ans (A)

$$\lim_{t \rightarrow 1^-} (1-t) \sum \underbrace{\{t^r - t^{2r} + t^{3r} - t^{4r} + \dots \text{upto } \infty\}}_{\text{G.P.}}$$

$$= \lim_{t \rightarrow 1^-} (1-t) \left[\underbrace{\sum t^r}_{\text{G.P.}} - \underbrace{\sum t^{2r}}_{\text{G.P.}} + \underbrace{\sum t^{3r}}_{\text{G.P.}} - \underbrace{\sum t^{4r}}_{\text{G.P.}} + \dots \text{upto } \infty \right]$$

$$= \lim_{t \rightarrow 1^-} (1-t) \left[\frac{t}{1-t} - \frac{t^2}{1-t^2} + \frac{t^3}{1-t^3} - \frac{t^4}{1-t^4} + \dots \text{upto } \infty \right]$$

$$= \lim_{t \rightarrow 1^-} \left[t - \frac{t^2}{1+t} + \frac{t^3}{1+t+t^2} - \dots \text{upto } \infty \right]$$

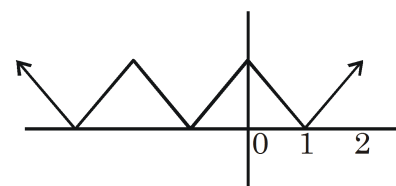
$$= 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots \text{upto } \infty$$

$$= \ln 2$$

5. Ans (B,C,D)

Check $f(-x) = f(x)$

In option (C), we have



function is periodic, from figure it is an even function.

6. Ans (B,C)

$$\pi - \cos^{-1} \left(\frac{1-x^2}{1+x^2} \right) + \sin^{-1} \left(\frac{2x}{1+x^2} \right) - \tan^{-1} \left(\frac{2x}{1-x^2} \right) = \frac{2\pi}{3}$$

$$\Rightarrow \cos^{-1}(\cos 2\theta) + \tan^{-1}(\tan 2\theta) - \sin^{-1}(\sin 2\theta) = \frac{\pi}{3}$$

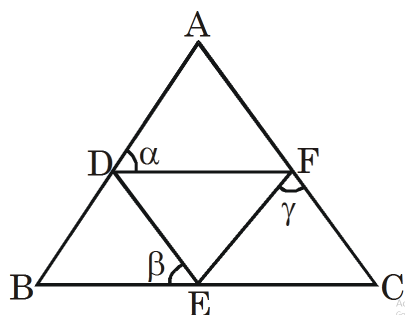
$$\theta \in \left(0, \frac{\pi}{4}\right) \Rightarrow 2\theta = \frac{\pi}{3} \Rightarrow \theta = \frac{\pi}{6} \Rightarrow x = \frac{1}{\sqrt{3}}$$

$$\theta \in \left(\frac{\pi}{4}, \frac{\pi}{2}\right) \Rightarrow \theta = \frac{7\pi}{18} \Rightarrow x = \tan\left(\frac{7\pi}{18}\right) = \tan 70^\circ$$

$$\theta \in \left(-\frac{\pi}{4}, 0\right) \Rightarrow \theta = -\frac{\pi}{6} \Rightarrow x = \left(\frac{-1}{\sqrt{3}}\right)$$

no. of solution is 3.

7. Ans (A)



Let $\angle ABC = \angle ACB = \theta$

$$\text{Now, } \alpha + 60^\circ = \theta + \beta \quad \dots(1)$$

$$\text{and } \beta + 60^\circ = \theta + \gamma \quad \dots(2)$$

$$\alpha - \beta = \beta - \gamma$$

$$\beta = \frac{\alpha + \gamma}{2}$$

8. Ans (A,B,C,D)

$$x^{12} - 1 = (x-1)(x^3+1)(x^6+1)(x^2+x+1)$$

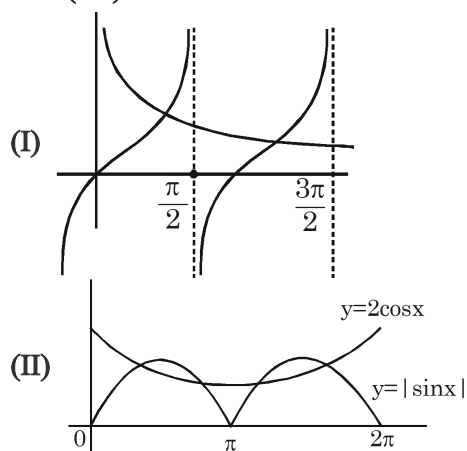
$$x^{12} = 20 \left(x + \frac{1}{x} + 1 \right) + 1$$

$$\text{Hence } x^{12} > 61$$

PART-3 : MATHEMATICS

SECTION-I (ii)

9. Ans (D)

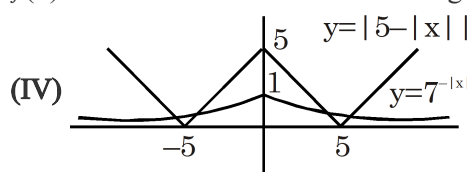


(III) $f(x) = 0$ has 8 real roots

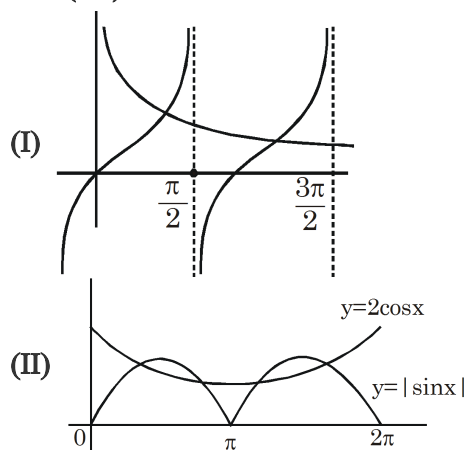
$\Rightarrow f(x) = 0$ has 4 positive roots.

$\Rightarrow f(x)$ can not have even no. of real roots so

$f(x) = 0$ has all roots real and one negative



10. Ans (A)

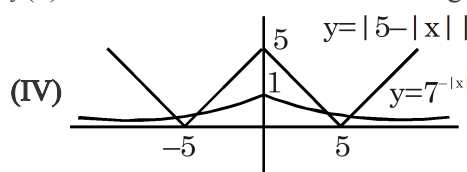


(III) $f(x) = 0$ has 8 real roots

$\Rightarrow f(x) = 0$ has 4 positive roots.

$\Rightarrow f(x)$ can not have even no. of real roots so

$f(x) = 0$ has all roots real and one negative



11. Ans (B)

(I) $4(\sin^6\theta + \cos^6\theta) = 10 - 18\sin\theta\cos\theta$

$$4\left(1 - \frac{3}{4}(\sin 2\theta)^2\right) = 10 - 9\sin 2\theta$$

$$4 - 3\sin^2 2\theta = 10 - 9\sin 2\theta$$

$$3\sin^2 2\theta - 9\sin 2\theta + 6 = 0$$

$$\sin^2 2\theta - 3\sin 2\theta + 2 = 0$$

$$(\sin 2\theta - 1)(\sin 2\theta - 2) = 0$$

$$\sin 2\theta = 1 \text{ in } [0, 3\pi]$$

$$\theta = \frac{\pi}{4}, \frac{5\pi}{4}, \frac{9\pi}{4}$$

(II) $x^2 - 6x + 10 = \sin^2 y$

$$\text{or } \sin^2 y = (x - 3)^2 + 1$$

$$\sin^2 y = 1 \text{ for } x = 3$$

$$\left(3, \frac{\pi}{2}\right) \text{ and } \left(3, \frac{3\pi}{2}\right)$$

(III) $\sin x < \frac{1}{2}$

$$x \in \left[0, \frac{\pi}{6}\right) \cup \left(\frac{5\pi}{6}, 2\pi\right] \quad \therefore x = 0, 3, 4, 5, 6$$

(IV) $\tan x + \sec x = 2\cos x$

$$1 + \sin x = 2\cos^2 x$$

$$1 + \sin x = 2 - 2\sin^2 x$$

$$2\sin^2 x + \sin x - 1 = 0$$

$$2\sin^2 x + 2\sin x - 1 = 0$$

$$(2\sin x - 1)(\sin x + 1) = 0$$

$$\sin x = \frac{1}{2}, \sin x = -1 \text{ (rejected)}$$

$$\sin x = \frac{1}{2} \text{ in } [0, 4\pi] \text{ no of solutions is 4.}$$

12. Ans (D)

(I) $4(\sin^6\theta + \cos^6\theta) = 10 - 18\sin\theta\cos\theta$

$$4\left(1 - \frac{3}{4}(\sin 2\theta)^2\right) = 10 - 9\sin 2\theta$$

$$4 - 3\sin^2 2\theta = 10 - 9\sin 2\theta$$

$$3\sin^2 2\theta - 9\sin 2\theta + 6 = 0$$

$$\sin^2 2\theta - 3\sin 2\theta + 2 = 0$$

$$(\sin 2\theta - 1)(\sin 2\theta - 2) = 0$$

$$\sin 2\theta = 1 \text{ in } [0, 3\pi]$$

$$\theta = \frac{\pi}{4}, \frac{5\pi}{4}, \frac{9\pi}{4}$$

(II) $x^2 - 6x + 10 = \sin^2 y$

$$\text{or } \sin^2 y = (x - 3)^2 + 1$$

$$\sin^2 y = 1 \text{ for } x = 3$$

$$\left(3, \frac{\pi}{2}\right) \text{ and } \left(3, \frac{3\pi}{2}\right)$$

(III) $\sin x < \frac{1}{2}$

$$x \in \left[0, \frac{\pi}{6}\right) \cup \left(\frac{5\pi}{6}, 2\pi\right] \quad \therefore x = 0, 3, 4, 5, 6$$

(IV) $\tan x + \sec x = 2\cos x$

$$1 + \sin x = 2\cos^2 x$$

$$1 + \sin x = 2 - 2\sin^2 x$$

$$2\sin^2 x + \sin x - 1 = 0$$

$$2\sin^2 x + 2\sin x - 1 = 0$$

$$(2\sin x - 1)(\sin x + 1) = 0$$

$$\sin x = \frac{1}{2}, \sin x = -1 \text{ (rejected)}$$

$$\sin x = \frac{1}{2} \text{ in } [0, 4\pi] \text{ no of solutions is 4.}$$

PART-3 : MATHEMATICS

SECTION-II

1. Ans (0.14)

$$S_n = \sum_{k=0}^n \tan^{-1} \frac{49}{49k^2 + 63k + 57}$$

$$= \sum_{k=0}^n \tan^{-1} \frac{1}{k^2 + \frac{63}{49}k + \frac{57}{49}}$$

$$= \sum_{k=0}^n \tan^{-1} \frac{1}{1 + \left(k + \frac{8}{7}\right) \left(k + \frac{1}{7}\right)}$$

$$= \sum_{k=0}^n \left(\tan^{-1} \left(k + \frac{8}{7}\right) - \tan^{-1} \left(k + \frac{1}{7}\right) \right)$$

$$\Rightarrow S_n = \tan^{-1} \left(n + \frac{8}{7}\right) - \tan^{-1} \frac{1}{7}$$

$$S_\infty = \cot^{-1} \frac{1}{7}$$

2. Ans (8.33)

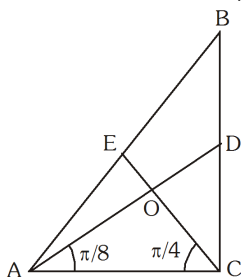
Let O be the point of intersection of the medians of triangle ABC. Then the area of $\triangle ABC$ is three times of that $\triangle AOC$. Now, in $\triangle AOC$, $AO = \frac{10}{3}$

Therefore, applying the sine rule to $\triangle AOC$, we get

$$\frac{OC}{\sin(\frac{\pi}{8})} = \frac{AO}{\sin(\frac{\pi}{4})} \Rightarrow OC = \frac{10}{3}, \frac{\sin(\frac{\pi}{8})}{\sin(\frac{\pi}{4})}$$

area of $\triangle AOC =$

$$\frac{1}{2} AO \cdot OC \cdot \sin \angle AOC$$



$$= \frac{1}{2} \cdot \frac{10}{3} \cdot \frac{10}{3} \cdot \frac{\sin(\frac{\pi}{8})}{\sin(\frac{\pi}{4})} \cdot \sin\left(\frac{\pi}{2} - \frac{\pi}{8}\right)$$

$$= \frac{50}{9} \cdot \frac{\sin(\frac{\pi}{8}) \cos(\frac{\pi}{8})}{\sin(\frac{\pi}{4})} = \frac{50}{18} = \frac{25}{9}$$

$$\therefore \text{area of } \triangle ABC = 3 \cdot \frac{25}{9} = \frac{25}{3}$$

3. Ans (2.00)

$$\lim_{x \rightarrow 0^+} \left[\frac{x}{\pi} \right] = \lim_{h \rightarrow 0} \left[\frac{h}{\pi} \right] = 0$$

$$\lim_{x \rightarrow 0^+} \frac{\sin^2 x}{\left[\frac{x}{\pi} \right] + \frac{x^2}{\pi^2}} \times \frac{\sin(\sin x) - \sin x}{ax^5 + bx^3 + c} = -\frac{\pi^2}{12}$$

$$x = 0 + h$$

$$\lim_{h \rightarrow 0} \frac{\sin^2 h}{\frac{h^2}{\pi^2}} \times \frac{2 \cos\left(\frac{\sinh+h}{2}\right) \sin\left(\frac{\sinh-h}{2}\right)}{ah^5 + bh^3 + c} = -\frac{\pi^2}{12}$$

$$\lim_{h \rightarrow 0} \frac{\pi^2 \cdot \sin^2 h}{h^2} \times 2 \cos\left(\frac{\sinh+h}{2}\right)$$

$$\frac{\sin\left(\frac{\sinh-h}{2}\right)}{\frac{\sinh-h}{2}} \times \frac{\frac{\sinh-h}{2}}{ah^5 + bh^3 + c} = -\frac{\pi^2}{12}$$

$$\Rightarrow \lim_{h \rightarrow 0} \frac{\sinh-h}{ah^5 + bh^3 + c} = -\frac{1}{12}$$

$$\lim_{h \rightarrow 0} \frac{\sinh-h}{h^3 (ah^2 + b)} = -\frac{1}{12} \quad (c=0)$$

$$-\frac{1}{6(b)} = -\frac{1}{12} \therefore b = 2$$

$$a \in \mathbb{R}, b = 2, c = 0$$

$$b + c = 2 + 0 = 2$$

4. Ans (0.25)

$$\lim_{x \rightarrow \infty} x^2 \tan\left(\ln\left(\cos \frac{\pi}{x}\right)^{1/2}\right)$$

$$= \lim_{x \rightarrow \infty} x^2 \left(\ln\left(\cos \frac{\pi}{x}\right)^{1/2}\right)$$

$$= \lim_{x \rightarrow \infty} \frac{x^2}{2} \ln\left(\cos \frac{\pi}{x}\right)$$

$$= \lim_{x \rightarrow \infty} \frac{x^2}{2} \left(\ln\left(1 + \left(\cos \frac{\pi}{x} - 1\right)\right)\right)$$

$$= \lim_{x \rightarrow \infty} \frac{x^2}{2} \left(\cos \frac{\pi}{x} - 1\right)$$

$$= \lim_{x \rightarrow \infty} \frac{x^2}{2} \left(-2\sin^2 \frac{\pi}{2x}\right) = -\frac{\pi^2}{4}$$

5. Ans (2100.00)

$$\lim_{\theta \rightarrow \frac{\pi}{2}^-} \frac{1}{\tan^9 \theta} \left((1 + \tan \theta)^{10} + (2 + \tan \theta)^{10} + \dots + (20 + \tan \theta)^{10} \right) - 20 \tan \theta$$

$$\lim_{\theta \rightarrow \frac{\pi}{2}^-} \frac{(1 + \tan \theta)^{10} + \dots + (20 + \tan \theta)^{10} - 20 \tan^{10} \theta}{\tan^9 \theta}$$

$$\left((\tan^{10} \theta + {}^{10}C_1 \tan^9 \theta) + (\tan^{10} \theta + {}^{10}C_1 2 \tan^9 \theta) + \dots + (\tan^{10} \theta + {}^{10}C_1 20 \tan^9 \theta) \right) - 20 \tan^{10} \theta$$

$$\lim_{\theta \rightarrow \frac{\pi}{2}^-} \frac{\dots}{\tan^9 \theta}$$

$$\Rightarrow {}^{10}C_1 (1 + 2 + 3 + \dots + 20) = \frac{10 \cdot 20 \cdot 21}{2}$$

$$\Rightarrow 2100$$

6. Ans (0.00)

$$2 \sin 2x + \sin x (1 - 2 \cos 2x) = 4$$

$$4 \sin x \cos x + \sin x (1 - 2(2 \cos^2 x - 1)) = 4$$

$$4 \cos x + 1 - 4 \cos^2 x + 2 = \frac{4}{\sin x}$$

$$- [4 \cos^2 x - 4 \cos x] + 3 = \frac{4}{\sin x}$$

$$- [(2 \cos x)^2 - 1] + 3 = \frac{4}{\sin x}$$

$$4 - (2 \cos x - 1)^2 = \frac{4}{\sin x}$$

$$\Rightarrow \text{L.H.S.} \leq 4 \text{ and R.H.S.} \geq 4$$

$$\text{and equality holds when } \cos x = \frac{1}{2} \text{ and } \sin x = 1$$

which is not possible.

$$\Rightarrow \text{No solution}$$

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : TOAS, TNAS & TAAS)
ANSWER KEY
PAPER-2
PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A	A	A	D	C	B	A	B	B	A
	Q.	11	12	13	14	15	16	17	18	19	20
SECTION-II	A.	B	B	B	B	B	B	A	B	B	B
	Q.	1	2	3	4	5	6	7	8	9	10
	A.	16	48	6	5	4	15	3	40	10	4

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	C	B	D	A	B	B	C	A	B
	Q.	11	12	13	14	15	16	17	18	19	20
SECTION-II	A.	A	B	C	C	C	A	B	A	C	C
	Q.	1	2	3	4	5	6	7	8	9	10
	A.	5	4	3	4	384	30	8	0	3	4

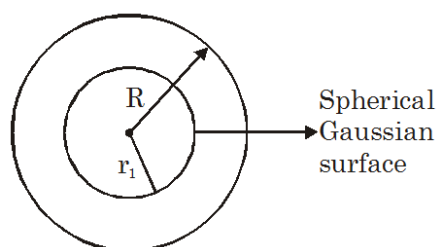
PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	A	B	D	B	A	A	D	D	A
	Q.	11	12	13	14	15	16	17	18	19	20
SECTION-II	A.	B	A	C	C	C	D	C	C	A	B
	Q.	1	2	3	4	5	6	7	8	9	10
	A.	7	2	1	3	0	6	8	786	2	7

HINT – SHEET
PART-1 : PHYSICS
SECTION-I

 1. **Ans (A)**

$$\oint \vec{E} \cdot d\vec{s} = \frac{q_{in}}{\epsilon_0}$$



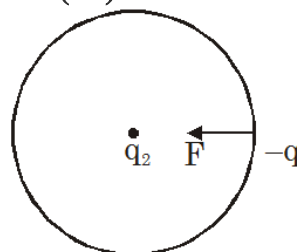
$$E \cdot 4\pi r_1^2 = \frac{q_{in}}{\epsilon_0}$$

$$E = \frac{q_{in}}{4\pi\epsilon_0 r_1^2}$$

$$\text{Where } q_{in} = \int_0^{r_1} \rho dV = \int_0^{r_1} \rho_0 \left(1 - \frac{r}{R}\right) 4\pi r^2 dr$$

$$= 4\pi\rho_0 \left(\frac{r_1^3}{3} - \frac{r_1^4}{4R} \right)$$

$$\therefore \text{From (i) } E = \frac{\rho_0 r_1}{3\epsilon_0} \left(1 - \frac{3r_1}{4R} \right)$$

 2. **Ans (A)**


$$F = F_C$$

$$\frac{kq_1q_2}{r^2} = \frac{mv^2}{r} \Rightarrow v = \sqrt{\frac{q_1q_2}{4\pi\epsilon_0 rm}}$$

$$T = \frac{2\pi r}{v} = 2\pi r \sqrt{\frac{4\pi\epsilon_0 m}{q_1q_2}}$$

$$T = \sqrt{\frac{16\pi^3 m\epsilon_0 r^3}{q_1q_2}}$$

3. **Ans (A)**

$$\phi_1 = -E_0 (4a^2) \times a^2 = -E_0 a^4$$

$$\phi_2 = E_0 (9a^2) \times a^2 = E_0 9a^4$$

$$\text{Total flux } \phi = 5E_0 a^4; \frac{q}{\epsilon_0} = 5E_0 a^4$$

$$q = 5\epsilon_0 a^4 E_0$$

4. **Ans (D)**

$$x = 10 \sin \left(4t - \frac{\pi}{3} \right)$$

$$\Rightarrow 6 = 10 \sin \left(4t - \frac{\pi}{3} \right)$$

$$\Rightarrow \left(4t - \frac{\pi}{3} \right) = 37^\circ$$

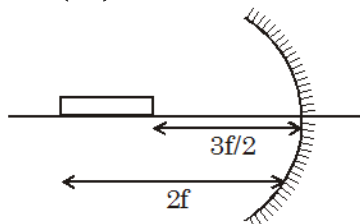
$$\therefore v = 10 \cos \left(4t - \frac{\pi}{3} \right) \times 4$$

$$= 10 \times \frac{4}{5} \times 4 = 32 \text{ m/s}$$

5. **Ans (C)**

Field inside spherical cavity is uniform.

6. **Ans (B)**



$$u = -\frac{3f}{2}$$

$$f = -f$$

$$\frac{1}{v_1} + \frac{1}{u} = \frac{1}{f}$$

$$v_1 = -3f$$

$$v_2 = -2f$$

$$\text{Image length} = |-3f| - |-2f| = f$$

8. **Ans (B)**

Let x_1 and x_2 be elongations in lower and upper spring

$$\Rightarrow \frac{-x_1 + x}{2} = x_2 \Rightarrow x = 2x_2 + x_1$$

Let T : tension in the string connecting the block.

$$\Rightarrow T = kx_1 \text{ and } T + kx_1 = kx_2$$

$$\Rightarrow 2x_1 = x_2$$

$$\Rightarrow x_1 = x/5 \text{ and } x_2 = 2x/5$$

Now loss in GPE of block = gain in KE + gain in EPEs

$$\Rightarrow mgx = \frac{1}{2}mv^2 + \frac{1}{2}kx_1^2 + \frac{1}{2}kx_2^2$$

$$\Rightarrow \frac{1}{2}mv^2 = mgx - \frac{1}{2}k \left\{ \frac{x^2}{25} + \frac{4x^2}{25} \right\}$$

$$= \frac{10mgx - kx^2}{10}$$

9. **Ans (B)**

$$f_{\max} = T_{\max}$$

$$4\mu mg = mg + \frac{mv_{\max}^2}{\ell}; v_{\max} = A\sqrt{\frac{g}{\ell}}$$

$$\Rightarrow \mu = \frac{1}{4} \left(1 + \frac{A^2}{\ell^2} \right)$$

12. **Ans (B)**

$$W = \int 2y dx + \int x^2 dy \text{ here } y = x$$

$$W = \int_0^5 2x dx + \int x^2 dx$$

$$W = 25 + \frac{125}{3} = \frac{200}{3} = 66.7J$$

14. **Ans (B)**

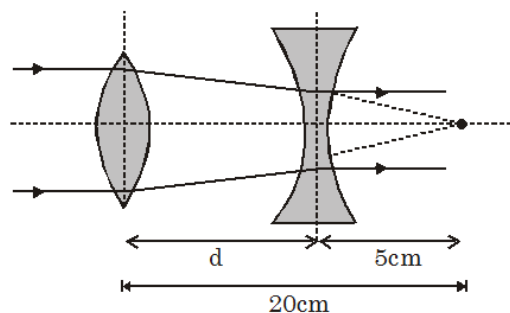
$$\vec{F}_C = - \left[\frac{\partial U}{\partial x} \hat{i} + \frac{\partial U}{\partial y} \hat{j} + \frac{\partial U}{\partial z} \hat{k} \right]$$

$$\text{Put } U = \frac{20yz}{x}$$

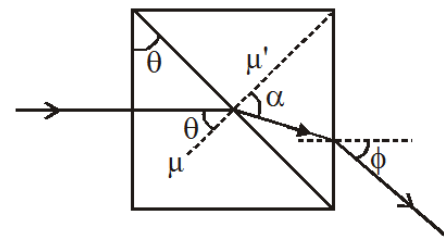
$$\vec{F}_C = \left(\frac{20yz}{x^2} \right) \hat{i} - \left(\frac{20z}{x} \right) \hat{j} - \left(\frac{20y}{x} \right) \hat{k}$$

15. **Ans (B)**

From figure it is clear that separation between lenses $d = 20 - 5 = 15 \text{ cm}$



17. **Ans (A)**



$$\mu \sin \theta = \mu' \sin \alpha$$

$$\Rightarrow \mu \theta = \mu' \alpha \dots (i)$$

$$\mu \sin(\alpha - \theta) = \sin \phi$$

$$\Rightarrow \mu'(\alpha - \theta) = \phi \dots (ii)$$

From (i) and (ii)

$$\mu \theta = \mu' \theta = \phi$$

$$\Rightarrow (\mu - \mu') = \frac{\phi}{\theta}$$

18. **Ans (B)**

For prism use have,

$$Li + Le = LA + LD \dots(i)$$

if $i = 15^\circ$, then $e = 60^\circ$

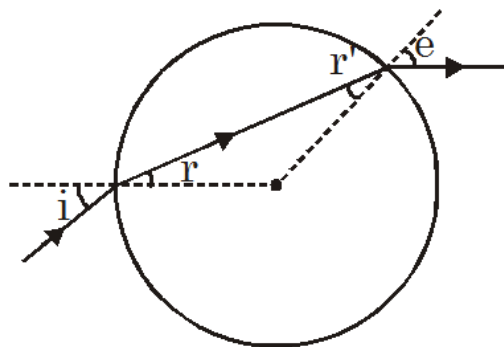
and $D = 30^\circ$

From eq. (i),

$$15^\circ + 60^\circ = A + 30^\circ$$

$$\Rightarrow A = 45^\circ$$

19. **Ans (B)**



$$\because r = r'$$

$$\therefore e = i = 60^\circ$$

$$\text{and } 1 \times \sin 60^\circ = \sqrt{3} \sin r$$

$$\Rightarrow r = 30^\circ$$

$$\therefore \delta = (60^\circ - 30^\circ) + (60^\circ - 30^\circ) = 60^\circ$$

20. **Ans (B)**

$$\ell = f_o + f_e, m = \frac{f_o}{f_e}$$

$$\Rightarrow 10 = \frac{20}{f_e} \Rightarrow f_e = 2 \text{ cm}$$

$$\therefore \ell = 20 + 2 = 22 \text{ cm.}$$

PART-1 : PHYSICS

SECTION-II

1. **Ans (16)**

Charge enclosed

$$q = \int_0^{R/2} Kr4\pi r^2 dr = \frac{4\pi K}{4} (R/2)^4$$

$$q = \frac{\pi K R^4}{16}$$

$$\text{Using gauss law ; } E4\pi(R/2)^2 = \frac{\pi K R^4}{16\epsilon_0}$$

$$E = \frac{KR^2}{16\epsilon_0} \text{ So, } N = 16$$

2. **Ans (48)**

$$v_1 = \frac{20 \times 30}{(20 - 30)} = -60 \text{ cm}$$

$$u \text{ for mirror} = 60 + 10 = 70 \text{ cm}$$

$$v \text{ for mirror} = 70 \text{ cm}$$

$$\therefore u \text{ for lens (again)} = 70 + 10 = 80 \text{ cm}$$

$$\therefore v_2 = \frac{80 \times 30}{(80 - 30)} = 48 \text{ cm}$$

3. **Ans (6)**

$$v = a\omega$$

$$0.6 = a \times 10$$

$$a = 0.06 \text{ m}$$

4. **Ans (5)**

$$K \frac{20}{100} = \frac{K10}{100} + mg$$

$$K = 10 \text{ mg}$$

$$T = 2\pi \sqrt{\frac{m}{K_1 + K_2}}$$

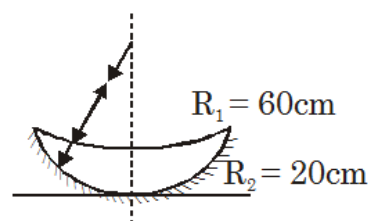
$$T = 2\pi \sqrt{\frac{m}{2K}} = 2\pi \sqrt{\frac{3}{20mg}}$$

$$T = \frac{\pi}{5} \sqrt{\frac{1}{2}} \Rightarrow N = 5$$

5. **Ans (4)**

Use refraction formula at both surfaces

6. **Ans (15)**



This silvered concavo-convex lens behaves like a mirror whose focal length can be calculated by the formula

$$\frac{1}{f} = \frac{2}{f_1} + \frac{1}{f_2}$$

f_1 = focal length of concave surface.

f_2 = focal length of concave mirror.

$$\therefore \frac{1}{f} = \frac{2}{-30} + \frac{1}{-10} = \frac{4}{30}$$

$\therefore f = -7.5 \text{ cm}$ Using mirror formula

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u} \Rightarrow \frac{1}{-7.5} = \frac{1}{-x} + \frac{1}{-x} \Rightarrow x = 15 \text{ cm}$$

8. Ans (40)

$$\frac{D - U}{-U} = -2$$

$$D - U = 2U$$

$$D = 3U$$

$$U = 20$$

$$m = \frac{F}{F - 20}$$

$$-2 = \frac{F}{F - 20}$$

$$-2F + 40 = F$$

$$F = \frac{40}{3} \text{ cm}$$

9. Ans (10)

$$W = W_{OA} + W_{AB}$$

$$= \int_0^1 x^2 dx + \int_0^2 y dy = \frac{1}{3} + \frac{4}{2} = \frac{7}{3} \text{ J}$$

$$\Rightarrow p + q = 10$$

10. Ans (4)

$$T = 2\pi \sqrt{\frac{1}{2mgd}}$$

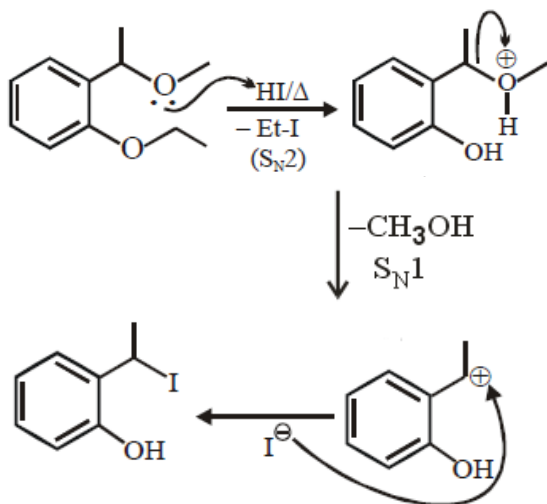
$$= 2\pi \sqrt{\frac{mR^2 + mR^2 + 4mR^2}{2mg \cdot \frac{3}{2}R}}$$

$$= 2\pi \sqrt{\frac{2R}{g}} = 4 \text{ sec}$$

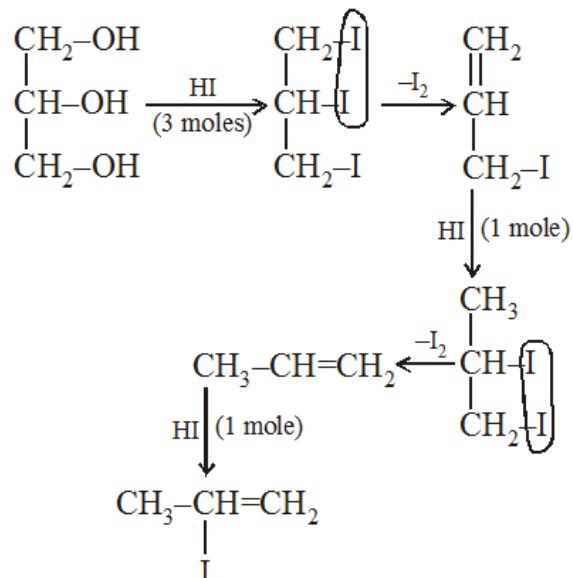
PART-2 : CHEMISTRY

SECTION-I

2. Ans (C)



6. Ans (B)



9. Ans (A)

$$\frac{P_A^0}{P_B^0} = \frac{1}{3}$$

$$\frac{P_A^l}{n_B^l} = \frac{1}{3} \quad y_A = ?$$

$$\therefore X_A = \frac{n_A^l}{n_A^l + n_B^l} = \frac{1}{1 + 3} = 0.25$$

$$Y_A = \frac{P_A^0 X_A}{P_A^0 X_A + P_B^0 X_B} = \frac{X_A}{X_A + \frac{P_B^0}{P_A^0} X_B}$$

$$\text{or, } Y_A = \frac{0.25}{0.25 + \frac{1}{3} + 0.75} = \frac{0.25}{0.25(1 + 9)} = \frac{1}{10}$$

$$= 0.1$$

10. Ans (B)

For hydrolysis of ester

$$r = k[H^+] [\text{ester}]$$

if 0.05 M HCl,

$$k_{HCl} = k = 0.05 \dots (1)$$

if 0.05 M H₂SO₄

$$k_{H_2SO_4} = 1 \times 0.1 \dots (2)$$

$$\frac{(2)}{(1)} \Rightarrow \frac{k_{H_2SO_4}}{k_{HCl}} = \frac{0.1}{0.05} = 2$$

$$\text{clearly, } k_{H_2SO_4} = 2k_{HCl}$$

11. Ans (A)

$$\text{Oleum} = 178\text{g}, \left(100 + \frac{900}{89}\right)\%$$

$$\text{for 100 g oleum water required} = \frac{900}{89}\text{g}$$

$$\therefore 178\text{ g oleum water required} = \frac{900}{89} \times \frac{178}{100}\text{g}$$

$$= 18\text{ g} = 1\text{ mole}$$

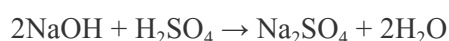


$\therefore$ In 178 g oleum,

$$\text{SO}_3 = 1\text{ mole} = 80\text{ g}$$

$$\text{H}_2\text{SO}_4 = 178 - 80 = 98\text{ g} = 1\text{ mole}$$

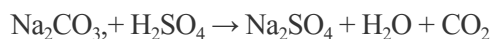
$$\therefore \text{Total H}_2\text{SO}_4 = 1 + 1 = 2\text{ mole}$$



2mole

$$\therefore \text{NaOH required} = 4\text{ mole.}$$

12. Ans (B)



$$106\text{ g} \quad \quad \quad 0.01\text{mole}$$

$$= \frac{1.06}{106}\text{mole} \quad \quad \quad = 0.01 \times 24.6$$

$$= 0.01\text{ mole} \quad \quad \quad = 0.246$$

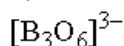
13. Ans (C)



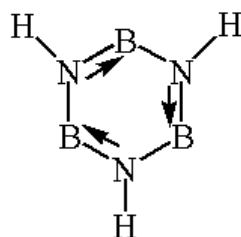
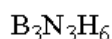
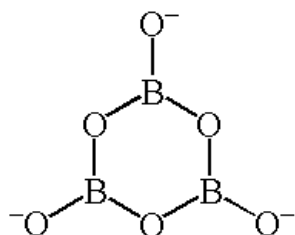
14. Ans (C)

(A) CHCl_3 is more acidic than CHF_3 as CCl_3^-

is stabilised by back bonding.



(C)



Both are planar

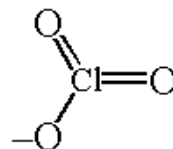
15. Ans (C)

(A) NO_2 odd e^- species

(B) BF_3 hypovalent

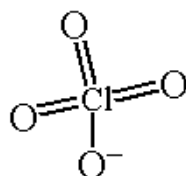
PF_5 hypervalent

(C) ClO_3^-



$$\text{BO} = 1 + \frac{2}{3}$$

$$= \frac{5}{3} = 1.66$$

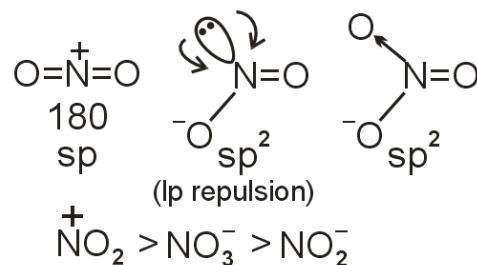


$$\text{BO} = 1 + \frac{3}{4}$$

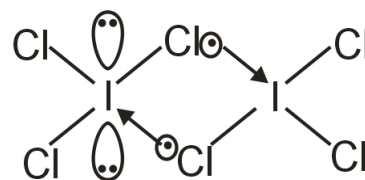
$$= \frac{7}{4} = 1.75$$

16. Ans (A)

(A) $\text{BF}_3 < \text{BCl}_3 < \text{BBR}_3$ [Lewis acidic order]



17. Ans (B)



(B) $\text{sp}^3\text{d}^2 - \text{sp}^3 - \text{sp}^3\text{d}^2$

18. Ans (A)

$$\sigma_{1s}^2 \quad \sigma_{1s}^{*2} \quad \sigma_{2s}^2 \quad \sigma_{2s}^{*2} \quad \pi_{2py}^2 \quad \pi_{2pz}^2 \quad 2\pi \text{ bond}$$

PART-2 : CHEMISTRY

SECTION-II

4. Ans (4)

Number of moles of cation vacancy

$$= \frac{6.02 \times 10^{21}}{6.02 \times 10^{23}} = 10^{-2} \text{ mole}$$

Number of moles of M^{+2} ions = 10^{-2} mole

Number of moles of missing y^- ions

= 2×10^{-2} mole % decreases in density

$$= \frac{30 \times 10^{-2} - 20 \times 10^{-2}}{25} \times 100\% = 0.4\% = x\%$$

$$\Rightarrow 10x = 4$$

5. Ans (384)

585.5 g solution, 0.5 molal sucrose $\Delta T_f = 4^\circ\text{C}$

1000 g solvent contains 0.5 mole sucrose

$$= 0.5 \times 342 = 171 \text{ g sucrose}$$

$$\therefore \text{mass of solution} = 1000 + 171 = 1171 \text{ g.}$$

$\therefore$ 1171 g solution contains 0.5 mole sucrose.

$$\therefore 585.5 \text{ solution contains } \frac{0.5}{1171} \times 585.5 = 0.25$$

mole sucrose.

$$\therefore \text{mass of solvent} = 585.5 - 0.25 \times 342 = 500 \text{ g}$$

Let x g ice is separated out

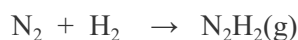
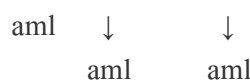
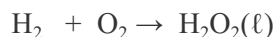
$$\Delta T_f = k_f m$$

$$\text{or, } 4 = \frac{1.86 \times 0.25 \times 1000}{(500 - x)}$$

$$\text{or, } 500 - x = 116.25$$

$$\text{or, } x = 383.75 \text{ g} = 384 \text{ (nearest integer)}$$

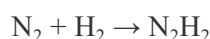
6. Ans (30)



$$\text{O}_2 = (a + 10) \text{ ml}$$

contraction in vol = 60

$$\text{or, } 2a + b = 60 \dots (1)$$



cml excess

$$1 \times C = 10$$

$$C \Rightarrow 10 \text{ ml}$$

$$\text{N}_2 = (b + 10) \text{ ml}$$

$$\text{H}_2 = (a + 10) \text{ ml}$$

$$\therefore a + b + b + 10 + a + 10 = 100$$

$$\text{or, } 2a + 2b = 80$$

$$\text{or, } a + b = 40 \text{ ml} \dots (2)$$

$$(1) + (2) \Rightarrow a = 20 \text{ ml} \therefore b = 20 \text{ ml}$$

$$\therefore \text{N}_2 = 30 \text{ ml}, \text{H}_2 = 40 \text{ ml}, \text{O}_2 = 30 \text{ ml}$$

$$\therefore \text{mole \% of N}_2 = \frac{30}{100} \times 100 = 30\%$$

7. Ans (8)

dipole moment = 1.6 D = μ_0

Bond length = 1.625 Å

$$\% \text{ ionic} = \frac{\mu_{\text{obs}}}{\mu_{\text{theo}}} \times 100 = \frac{1.6}{q \times d} \times 100$$

$$= \frac{1.6\text{D}}{(1.6 \times 10^{-19})(1.625 \times 10^{-10})} \times 100$$

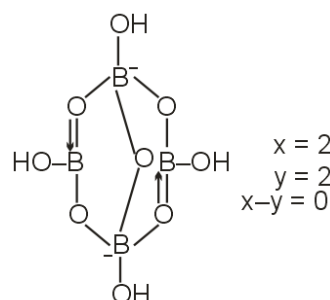
$$= \frac{1.6 \times 3.3 \times 10^{-30}}{(1.6 \times 10^{-19}) \times (1.625 \times 10^{-10})} \times 100$$

$$x = \frac{3.3}{1.625} \times 10 = \frac{33}{1.625} = 20.3$$

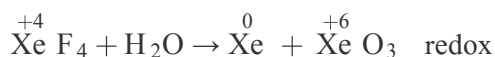
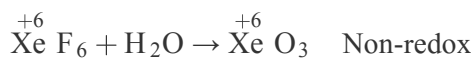
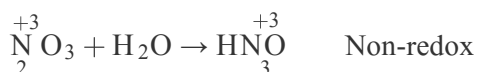
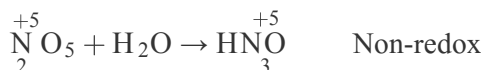
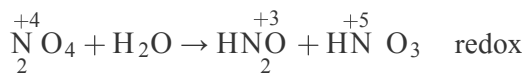
$$x \times 0.39$$

$$20.3 \times 0.39 \approx 8$$

8. Ans (0)



9. Ans (3)



PART-3 : MATHEMATICS

SECTION-I

1. Ans (C)

$$4f(x^{-1} + 1) + 8f(x + 1) = \log_{12}x \quad \dots(i)$$

$$8f\left(\frac{1}{x} + 1\right) + 8f(x + 1) = \log_{12}x \quad \dots(ii)$$

$$\text{Replace } x \rightarrow \frac{1}{x}$$

solving i & ii we get

$$f(x + 1) = \frac{\log_{12}x}{4} \quad \dots(iii)$$

Put $x = 9$; $x = 12$; $x = 16$ & add

$$\text{two get } f(10) + f(13) + f(17) = \frac{3}{4}$$

$$\Rightarrow 4(f(10) + f(13) + f(17)) = 3$$

2. Ans (A)

$\cot x > 1$ & 'tanx' is fractional

$$\Rightarrow t_4 > t_3 > t_1 > t_2$$

3. Ans (B)

$$f(x) = \frac{x+a}{x+b}$$

$$f^{-1}(x) = \frac{a-xb}{x-1}$$

Given $f(x) = f^{-1}(x)$

solving we get $b = -1$

$$\Rightarrow b + 2 = 1$$

4. Ans (D)

$$f(f(x)) = x$$

$$\Rightarrow \underbrace{f(f(f(\dots(x))))}_{\text{odd no. of times } f} = f(x) \text{ and } \underbrace{f(f(f(\dots(x))))}_{\text{even no. of times } f} = x$$

5. Ans (B)

$$A = \sin^{-1}x, B = \sin^{-1}y, C = \sin^{-1}z$$

A, B, C are angles of triangle.

$$\Rightarrow \sin 2A + \sin 2B + \sin 2C = 4 \sin A \sin B \sin C$$

6. Ans (A)

$$\tan^{-1}\left(\frac{x+1+\frac{1}{x-1}}{1-\frac{x+1}{x-1}}\right) = 2\tan^{-1}\frac{4}{3} = \tan^{-1}\left(-\frac{24}{7}\right)$$

$$\Rightarrow x^2 = \frac{48}{7}$$

$$x = 4\sqrt{\frac{3}{7}}$$

7. Ans (A)

$$(\tan^{-1}x)^2 + (\cot^{-1}x)^2 = \frac{5\pi^2}{8}; \tan^{-1}x = t$$

$$t^2 + \left(\frac{\pi}{2} - t\right)^2 = \frac{5\pi^2}{8} \Rightarrow t = -\frac{\pi}{4}, \frac{3\pi}{4}$$

$$\text{But } \tan^{-1}x \neq \frac{3\pi}{4} \Rightarrow \tan^{-1}x = -\frac{\pi}{4} \Rightarrow x = -1$$

$\Rightarrow$ one solution.

8. Ans (D)

$$\lim_{x \rightarrow 0} \frac{\ln(1+3x^2) \sin(\tan^{-1}x)}{(3^x - 1)(1 - \cos 2x)} \left[\frac{\tan x}{x} \right]$$

$$= \lim_{x \rightarrow 0} \frac{(3x^2)(x)}{(x \ln 3) \left(\frac{4x^2}{2}\right)} (1) = \frac{3}{2 \ln 3}$$

9. Ans (D)

$$\lim_{x \rightarrow 0} \frac{\alpha \sin x - \beta x \cos x}{\left(\frac{\tan x - \sin x}{x^3}\right) x^3} = 1$$

$$\lim_{x \rightarrow 0} \frac{\alpha \left(x - \frac{x^3}{6}\right) - \beta x \left(1 - \frac{x^2}{2}\right)}{x^3} = \frac{1}{2}$$

$$\lim_{x \rightarrow 0} \frac{(\alpha - \beta)x + \left(\frac{\beta}{2} - \frac{\alpha}{6}\right)x^3}{x^3} = \frac{1}{2}$$

$$\Rightarrow \alpha - \beta = 0 \text{ and } \frac{\beta}{2} - \frac{\alpha}{6} = \frac{1}{2} \Rightarrow \alpha = \beta = \frac{3}{2}$$

$$\Rightarrow \alpha + \beta = 3$$

10. Ans (A)

$$\lim_{x \rightarrow 0} \frac{(2x) \left(x + \frac{x^3}{3} - x + \frac{x^3}{3}\right)}{(e) \left(-\frac{(3x)^2}{2}\right) \left(-\frac{x^2}{2}\right)} = \frac{16}{27e}$$

11. **Ans (B)**

$$L = \lim_{x \rightarrow 0} \frac{\cos 2x - (1 + x \sin x)}{x \cdot \frac{x}{2} \cdot \frac{\tan \frac{x}{2}}{\frac{x}{2}}} \cdot \frac{1}{\sqrt{\cos 2x} + \sqrt{1 + x \sin x}}$$

$$= \lim_{x \rightarrow 0} \frac{\cos 2x - 1 - x \sin x}{x^2} = \lim_{x \rightarrow 0} \frac{-(x \sin x + 2 \sin^2 x)}{x^2}$$

$$\Rightarrow L = \lim_{x \rightarrow 0} - \left(\frac{\sin x}{x} + \frac{2 \sin^2 x}{x^2} \right) = -3$$

12. **Ans (A)**

$$\lim_{x \rightarrow 1} \frac{(x-1)(x^2+x+1)}{(x-1) \left\{ (x-2) - \frac{\sin(x-1)}{x-1} \right\}}$$

$$= \frac{3}{(-1-1)} = -\frac{3}{2}$$

13. **Ans (C)**

$$L = \lim_{x \rightarrow 0} \frac{\sin(2\pi \sec x - 2\pi)}{x^2}$$

$$= \lim_{x \rightarrow 0} \frac{\sin(2\pi(\sec x - 1))}{x^2}$$

$$L = \lim_{x \rightarrow 0} \underbrace{\frac{\sin(2\pi(\sec x - 1))}{2\pi(\sec x - 1)}}_1 \times \frac{2\pi(\sec x - 1)}{x^2}$$

$$= \lim_{x \rightarrow 0} \frac{2\pi(1 - \cos x)}{x^2 \cdot \cos x} = \pi$$

14. **Ans (C)**

$$y = f(x) = k \sin x + \cos x$$

$$\Rightarrow y \in \left[-\sqrt{k^2 + 1}, \sqrt{k^2 + 1} \right], y \in [-5, 5]$$

$$\Rightarrow k^2 + 1 = 25 \Rightarrow k = \pm\sqrt{24}$$

$$\Rightarrow \text{integer} \in \{-4, -3, -2, -1, 0, 1, 2, 3, 4\}$$

15. **Ans (C)**

$$\because a^2 + b^2 = 1 + 1 + 2\sin(A+B)$$

$$\Rightarrow \sin(A+B) = \frac{a^2 + b^2 - 2}{2}$$

16. **Ans (D)**

Given product

$$= \frac{\cos 16 \frac{\pi}{5}}{16 \cos \frac{\pi}{5}} = \frac{\cos(3\pi + \frac{\pi}{5})}{16 \cos \frac{\pi}{5}} = -\frac{1}{16}$$

17. **Ans (C)**

$$\tan \theta + \tan 2\theta + \tan 3\theta = \tan \theta \tan 2\theta \tan 3\theta$$

$$\Rightarrow \theta + 2\theta + 3\theta = n\pi$$

$$\Rightarrow 6\theta = n\pi, n \in \mathbb{I}$$

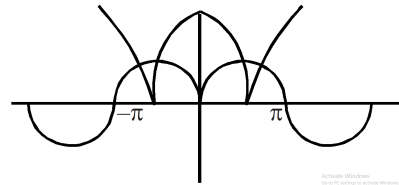
$$\text{In } (0, \pi) : \theta = \frac{\pi}{6}, \frac{\pi}{3}, \frac{\pi}{2}, \frac{2\pi}{3}, \frac{5\pi}{6}$$

$$\frac{\pi}{2}, \frac{\pi}{6}, \frac{5\pi}{6} \text{ are not in domain}$$

$$\therefore \theta = \frac{\pi}{3}, \frac{2\pi}{3} \text{ are the solutions}$$

18. **Ans (C)**

$$\sin |x| = 2|\cos x|$$



clearly 4 solutions.

19. **Ans (A)**

$$y - x^2 - 6x + 10 = (x-3)^2 + 1 \geq 1$$

$$\Rightarrow \sin x - \cos x = \frac{1}{\sqrt{2}}$$

$$\frac{1}{\sqrt{2}} \sin x - \frac{1}{\sqrt{2}} \cos x = \frac{1}{2}$$

$$\sin\left(x - \frac{\pi}{4}\right) = \sin \frac{\pi}{6}$$

$$x = \frac{\pi}{4} + n\pi + (-1)^n \frac{\pi}{6}, n \in \mathbb{I}$$

20. **Ans (B)**

$$\text{Area} = \frac{1}{2}(a+b)^2 = \frac{81}{2}$$

$$\Rightarrow a+b = 9 \Rightarrow a=6 \text{ \& } b=3$$

$$\Rightarrow c = \sqrt{6^2 + 3^2} = 3\sqrt{5}$$

PART-3 : MATHEMATICS

SECTION-II

1. **Ans (7)**

$$\cos^{-1} \sin(\cos^{-1} x) + \sin^{-1}(\cos(\sin^{-1} x))$$

$$= \cos^{-1} \sqrt{1-x^2} + \sin^{-1} \sqrt{1-x^2} = \frac{\pi}{2}$$

$$\Rightarrow f(x) = 1$$

$$\therefore \sum_{x=1}^7 f\left(\frac{x}{8}\right) = 7$$

2. **Ans (2)**

$$\begin{aligned}
 & \sin(\cot^{-1}(\cos \tan^{-1} x)) \\
 &= \sin\left(\cot^{-1} \cos \cos^{-1}\left(\frac{1}{\sqrt{1+x^2}}\right)\right) \\
 &= \sin\left(\cot^{-1}\left(\frac{1}{\sqrt{1+x^2}}\right)\right) \\
 &= \sin\left(\tan^{-1} \sqrt{1+x^2}\right) \\
 &= \sin\left(\sin^{-1}\left(\frac{\sqrt{1+x^2}}{\sqrt{2+x^2}}\right)\right) = \sqrt{\frac{1+x^2}{2+x^2}} \\
 &\Rightarrow \sqrt{\frac{1+x^2}{2+x^2}} = a \Rightarrow x^2 = \frac{2a^2-1}{1-a^2} \\
 &\Rightarrow a \neq \pm 1 \\
 &(1-a^2)(2a^2-1) \geq 0 \\
 &\Rightarrow \frac{1}{2} \leq a^2 < 1 \Rightarrow a \in \left[\frac{1}{\sqrt{2}}, 1\right) \\
 &p = \frac{1}{\sqrt{2}}, q = 1 \Rightarrow q + \frac{1}{p} = 1 + \sqrt{2} \\
 &\Rightarrow \left[q + \frac{1}{p}\right] = 2
 \end{aligned}$$

3. **Ans (1)**

$$\begin{aligned}
 & \tan^{-1}\left(\frac{1}{3}\right) + \tan^{-1}\left(\frac{2}{9}\right) + \tan^{-1}\left(\frac{4}{33}\right) + \dots \\
 & t_n = \tan^{-1}\left(\frac{2^{n-1}}{1+2^{2n-1}}\right) \\
 &= \tan^{-1}\left(\frac{2^n - 2^{n-1}}{1+2^n \cdot 2^{n-1}}\right) \\
 &= \tan^{-1}(2^n) - \tan^{-1}(2^{n-1})
 \end{aligned}$$

4. **Ans (3)**

$$\begin{aligned}
 & \tan^{-1}x + \tan^{-1}y + \tan^{-1}z = \frac{\pi}{2} \\
 & \text{only when } xy + yz + zx = 1 \quad \dots\dots(i) \\
 & \text{Also } x + y + z = \sqrt{3} \quad \dots\dots(ii) \\
 & \Rightarrow x^2 + y^2 + z^2 + 2(xy + yz + zx) = 3 \\
 & \Rightarrow x^2 + y^2 + z^2 = 1 \quad \dots\dots(iii) \\
 & \therefore (x-y)^2 = x^2 + y^2 - 2xy \\
 & \quad (y-z)^2 = y^2 + z^2 - 2yz \\
 & \quad (z-x)^2 = z^2 + x^2 - 2zx \\
 & \therefore (x-y)^2 + (y-z)^2 + (z-x)^2 = 0 \\
 & \Rightarrow x = y = z \\
 & \therefore x = y = z = \frac{1}{\sqrt{3}} \\
 & \therefore \frac{1}{xyz} = 3\sqrt{3}
 \end{aligned}$$

5. **Ans (0)**

$$\begin{aligned}
 & 2^{4x^2-1} - 2^{x^4} > 0 \\
 & 2^{4x^2-1} > 2^{x^4} \\
 & 4x^2 - 1 > x^4 \\
 & x^4 - 4x^2 + 1 < 0 \\
 & 2 - \sqrt{3} < x^2 < 2 + \sqrt{3} \\
 & (1) \quad x^2 > 2 - \sqrt{3} \\
 & \quad x^2 - \left(\frac{\sqrt{3}-1}{\sqrt{2}}\right)^2 > 0 \\
 & \quad x \in \left(-\infty, -\frac{(\sqrt{3}-1)}{\sqrt{2}}\right) \cup \left(\frac{\sqrt{3}-1}{\sqrt{2}}, \infty\right) \dots(i) \\
 & (2) \quad x^2 < 2 + \sqrt{3} \\
 & \quad x^2 - \left(\frac{\sqrt{3}+1}{\sqrt{2}}\right)^2 < 0 \\
 & \quad x \in \left(-\frac{(\sqrt{3}+1)}{\sqrt{2}}, \frac{\sqrt{3}+1}{\sqrt{2}}\right) \dots(ii) \\
 & \Rightarrow x \in \left(-\frac{\sqrt{3}+1}{\sqrt{2}}, -\frac{(\sqrt{3}-1)}{\sqrt{2}}\right) \cup \left(\frac{\sqrt{3}-1}{\sqrt{2}}, \frac{\sqrt{3}+1}{\sqrt{2}}\right) \\
 & \Rightarrow a + b + c + d = 0
 \end{aligned}$$

6. **Ans (6)**

$$\begin{aligned}
 & \lim_{x \rightarrow 0} \frac{x^2(x - \tan x)}{(x - \sin x)} \cdot \frac{(\sin^{-1}x)^2}{(\sin^{-1}x + x)(x - \sin^{-1}x)} \\
 & \lim_{x \rightarrow 0} \frac{x^2\left(x - x - \frac{x^3}{3}\right)}{\left(x - x + \frac{x^3}{6}\right)} \cdot \frac{(\sin^{-1}x)^2}{\left(2x + \frac{x^3}{6}\right)\left(x - x - \frac{x^3}{6}\right)} \\
 & \lim_{x \rightarrow 0} \frac{x^2 \cdot \left(-\frac{x^3}{3}\right) (\sin^{-1}x)^2}{\frac{x^3}{6} \cdot x \left(2 + \frac{x^2}{6}\right) \left(-\frac{x^3}{6}\right)} = \frac{-\frac{1}{3}}{\frac{1}{6} \cdot 2 \cdot -\frac{1}{6}} = 6
 \end{aligned}$$

7. Ans (8)

$$\lim_{x \rightarrow 0} \left(\frac{\sin x}{x} \right)^{\frac{\sin x}{x - \sin x}} = 1^\infty$$

$$\Rightarrow e^{\lim_{x \rightarrow 0} \frac{\sin x}{x - \sin x} \left(\frac{\sin x - x}{x} \right)} = e^{-1}$$

$$\lim_{x \rightarrow 0} (x)^{\frac{1}{1-x}} = 1^\infty$$

$$\Rightarrow e^{\lim_{x \rightarrow 0} \frac{1}{1-x} (x-1)} = e^{-1}$$

$$4e(2 \cdot e^{-1}) = 8$$

8. Ans (786)

$$a(a - c) = b^2 - c^2$$

$$\Rightarrow a^2 - ac = b^2 - c^2$$

$$\Rightarrow \frac{a^2 + c^2 - b^2}{2ac} = \frac{1}{2}$$

$$\Rightarrow \cos B = \frac{1}{2} \Rightarrow \cos^2 B = \frac{1}{4}$$

$$\sin^2(\pi \cos^2 B) = \frac{1}{2}$$

$$1572 \sin^2(\pi \cos^2 B) = 786$$

9. Ans (2)

$$\frac{1}{r} = \frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3}$$

$$\frac{1}{r} = \frac{1}{4} + \frac{1}{6} + \frac{1}{8}$$

$$r = \frac{24}{13}$$

10. Ans (7)

$$\cot \frac{A}{2} \cdot \cot \frac{B}{2}$$

$$= \sqrt{\frac{s(s-a)s(s-b)}{(s-c)(s-b)(s-a)(s-c)}} = \frac{s}{s-c}$$

$$\therefore 3(a+b) = 4c \Rightarrow 3(2s-c) = 4c$$

$$\Rightarrow 6s = 7c \Rightarrow s = \frac{7c}{6}$$

$$\therefore \cot \frac{A}{2} \cdot \cot \frac{B}{2} = \frac{\frac{7c}{6}}{\frac{7c}{6} - c} = 7$$